Topic 15 — Sustainable Development & Environmental Governance

Covers syllabus (click to expand)

Sustainable Development: SD | SDGs | SDGs 2030 | Objectives | Indicators | Dimensions | Brundtland Report | Natural capital (प्राकृतिक पूंजी) | Sustainable agriculture | NMSA **Environmental Governance:** Governance | Environmental Justice | Environmental Ethics | Green Economy (हरित अर्थव्यवस्था) | Circular Economy (चक्रीय अर्थव्यवस्था)

Sources baked in: Brundtland Report 1987, UN SDGs 2030, Rio 1992/Agenda 21, UNEP Green Economy 2011, NCERT, NITI Aayog SDG Index, UPPCS PYQs 2018–2025 **Weight:** ★★★ High — SD A/R traps, 17 SDGs, Brundtland vs Limits to Growth, LiFE, UP Planning Dept **Last verified:** August 2026

Current Affairs (this topic)

- SDG progress reports and LiFE / Mission LiFE headlines refresh this chapter's CA.
- Brundtland / Agenda 21 / Rio years stay fixed — do not relocate them into Paris-only answers.

15.1 Sustainable Development

Identity: Brundtland 1987 | Three pillars | Inter/intra-generational equity | Indicators | Natural capital

- **Sustainable development** means meeting the needs of the present **without compromising** the ability of future generations to meet their own needs.
- This definition comes from the **Brundtland Report (1987)** (ब्रंटलैंड रिपोर्ट), formally titled ***Our Common Future***, by the **World Commission on Environment and Development (WCED)** chaired by **Gro Harlem Brundtland**.
- The Brundtland Report is **not** *The Limits to Growth*. That 1972 study was by the **Club of Rome**.
- **Inter-generational equity** protects future generations. **Intra-generational equity** ensures fair sharing among people alive today.
- Sustainable development is a matter of **inter-generational sensibility** in the use of **natural resources**.
- The **three dimensions/pillars** are **economic** (viable growth), **social** (equity, health, education), and **environmental** (ecological integrity). Some frameworks add **institutional governance**.
- **Objectives** include fulfilling **basic needs**, respecting **ecological limits** and carrying capacity, integrating environment into economic planning, and ensuring **public participation**.
- Core practical objectives also include **poverty reduction**, **biodiversity conservation**, and **prevention and control of pollution**.

- Development is unsustainable if resource extraction exceeds regeneration or waste exceeds environmental absorption capacity.
- **Natural capital** includes all renewable and non-renewable environmental resources such as air, water, land, minerals, and forests. **Roads** and other purely human-made assets are **not** natural capital.
- SD road-mapping often uses information from **remote sensing**, **GIS**, statistical methods, and environmental information systems.
- **Indicators** include **HDI** (UNDP), **ecological footprint** (पारिस्थितिक पदचिह्न), **SDG Index**, **Environmental Performance Index (EPI)**, and **231 SDG indicators**.
- **NITI Aayog** (नीति आयोग) publishes the **SDG India Index** ranking states on SDG progress.
- The timeline runs **Stockholm 1972** (स्टॉकहोम सम्मेलन) → **Brundtland 1987** → **Rio Earth Summit and Agenda 21 (1992) (एजेंडा 21)** → **SDGs adopted 2015**.

PYQ — UPPCS Prelims 2025, Q51

(A) SD should occur without harming the environment or compromising future generations' needs. (R) Agenda 21 was signed by world leaders in 1995.

A. Both true; R not explanation

B. A false, R true

C. A true, R false

D. Both true; R explains A

Show answer

Ans: C — Agenda 21 = **Rio 1992**, not 1995.

PYQ — UPPCS Prelims 2024, Q51

(A) SD was popularised by the Brundtland Report. (R) The Brundtland Report is also known as *The Limits to Growth*.

A. Both true; R not explanation

B. A false, R true

C. Both true; R explains A

D. A true, R false

Show answer

Ans: D — Brundtland = *Our Common Future* (1987).

Pillar	Focus
Economic	Employment, innovation, infrastructure
Social	Poverty, health, education, equity

Pillar	Focus
Environmental	Pollution, biodiversity, climate
Institutional	Governance, rule of law, participation

15.2 SDGs & SDGs 2030

Identity: 17 goals | 2030 deadline | 2015 adoption | MDGs replaced | NITI coordination

- The **2030 Agenda for Sustainable Development** was adopted by all UN member states on **25 September 2015**.
- It sets **17 Sustainable Development Goals (SDGs) (सतत विकास लक्ष्य)** with **169 targets** and **231 indicators** to be achieved by **2030**.
- **SDGs replaced MDGs**, which had **8 goals (2000–2015)** and focused mainly on developing countries.
- **SDGs are universal** — they apply to all countries, including developed ones.
- **Leave No One Behind** requires progress to reach the poorest and most vulnerable groups.
- **NITI Aayog** coordinates SDG localisation in India. **MoSPI** maintains the National Indicator Framework.
- India's large population (जनसंख्या) share makes its SDG performance critical for global success.
- A global **SDG Index / Dashboard** ranks countries on progress toward the 17 goals. **Finland**, **Sweden**, and **Denmark** typically lead.
- India ranked about **116th** in the **2017** global SDG Index and about **112th** among **166** countries in the **2023** reading (रीडिंग).
- **NITI Aayog's SDG India Index** ranks states and UTs. **Kerala** has repeatedly held the **first** position (including the 2019–20 and 2020–21 editions).
- In the 2019–20 / 2020-style top-five set often tested, leading states include **Kerala**, **Himachal Pradesh**, **Tamil Nadu**, and **Andhra Pradesh**. **Gujarat** was **not** in that top-five list.

PYQ — UPPCS Prelims 2025, Q142

Goal 1 / Goal 3 / Goal 4 / Goal 6 matched to: (1) Clean Water (2) Quality Education (3) No Poverty (4) Good Health

Show answer

Ans: B (3 4 2 1) — Goal 1 = No Poverty; 3 = Health; 4 = Education; 6 = Water.

PYQ — UPPCS Prelims 2025, Q120

(A) India's success is critical for global SDG success. (R) India accounts for nearly one-sixth of world population.

Show answer

Ans: D — Both true; R explains A.

PYQ — UPPCS Prelims 2024, Q98

Essential for SDG 4 (Quality Education):

1. Free and compulsory education
2. School infrastructure and digital transformation
3. Expansion of agricultural programmes
4. Increasing investment in technology

Show answer

Ans: A (1 and 2) — Agricultural expansion is not core to SDG 4.

Goal	Official name
1	No Poverty
2	Zero Hunger
3	Good Health and Well-being
4	Quality Education
5	Gender Equality
6	Clean Water and Sanitation
7	Affordable and Clean Energy
8	Decent Work and Economic Growth
9	Industry, Innovation and Infrastructure
10	Reduced Inequalities
11	Sustainable Cities and Communities
12	Responsible Consumption and Production
13	Climate Action
14	Life Below Water
15	Life on Land
16	Peace, Justice and Strong Institutions
17	Partnerships for the Goals
Feature	MDGs
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Goals	8
Period	2000–2015
Scope	Mainly developing countries

15.3 Environmental Governance, Justice & Ethics

Identity: Multi-level governance | Rio principles | Bhopal-EJ | Anthropocentrism vs ecocentrism | LiFE

- **Environmental governance** is the system of rules, institutions, and processes through which environmental decisions are made and enforced.
- It operates at **global** (UNEP, treaties), **national** (MoEFCC, Parliament), **state** (SPCBs), and **local** (panchayats, municipalities) levels.
- **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is India's nodal ministry for environment, forest, and climate policy.
- **Rio principles** include **Precautionary principle**, **Polluter Pays**, **Public Trust doctrine**, and **inter-generational equity** — adopted in Indian judgments like *Vellore Citizens' Welfare Forum*.
- Constitutional anchors include **Article 48A** (state shall protect environment), **Article 51A(g)** (citizens' duty to protect environment), and **Article 21** (right to healthy environment).
- **NGT (2010)** is the specialised tribunal for environmental cases and EC appeals.
- **Agenda 21 (1992)** is a Rio blueprint for sustainable development planning, including local-level action.
- **Environmental justice** means fair treatment of all in environmental benefits and burdens — both **distributive** (who gets pollution) and **procedural** (who participates in decisions).
- The **Bhopal Gas Tragedy (1984)** exposed environmental injustice and led to the **EPA 1986** (पर्यावरण संरक्षण 1986).
- UP hotspots include **Kanpur-Unnao tanneries** and the **Singrauli–Sonbhadra (सोनभद्र) coal belt**.
- **Environmental ethics** asks what moral duties humans owe to nature.
- **Anthropocentrism** values nature mainly for human use.
- **Biocentrism** gives intrinsic worth to all living beings.
- **Ecocentrism** values whole ecosystems.
- **Aldo Leopold's Land Ethic (1949)** stresses preserving the integrity of the biotic community.
- **Gandhi's trusteeship** holds that resources are held in trust for society and future generations.
- The **LiFE Mission** promotes individual pro-planet behaviour. The idea was announced at **COP26 Glasgow (2021)** and launched in **June 2022**.

PYQ — UPPCS Prelims 2025, Q111

UP department preparing SDG National Indicator Framework Progress Report:

1. Finance 2. Education 3. Planning

Show answer

Ans: B (Only 3) — **Planning Department**.

PYQ — UPPCS Prelims 2025, Q139

1. LiFE was launched in June 2022.
2. The idea was proposed at COP-25 in Madrid.

Show answer

Ans: D (Only 1) — Idea at **COP26 Glasgow**, not COP-25.

Concept	Fact
Environmental justice	Fair distribution + participation
Anthropocentrism	Human-centred value
Biocentrism	All life has worth
Ecocentrism	Ecosystem (पारिस्थितिकी तंत्र) wholes have worth
Leopold	Land Ethic, 1949
Gandhi	Trusteeship, limit wants

15.4 Green Economy & Circular Economy

Identity: UNEP 2011 green growth | 3R/EPR | Linear vs circular | SDG 12

- The **green economy** (UNEP, 2011) aims for improved well-being and social equity while **reducing environmental risk and ecological scarcity**.
- It seeks **low-carbon, resource-efficient, socially inclusive** growth — decoupling GDP from environmental harm.
- India examples include **solar expansion**, **National Green Hydrogen Mission (2023)**, **Net Zero 2070**, and **LiFE Mission**.
- The **circular economy** replaces the linear **take-make-dispose** model with loops of **reduce, reuse, recycle, repair, and recovery**.
- In a circular system, **waste is treated as a design flaw**, not an afterthought.
- **Extended Producer Responsibility (EPR)** (विस्तारित उत्पादक दायित्व) under Plastic, E-Waste, Battery, and Tyre rules makes producers responsible for end-of-life collection.
- **Circular economy** focuses on **material flows and waste elimination**.
- **Green economy** is **broader** — it also covers renewable energy (नवीकरणीय ऊर्जा), biodiversity, jobs, and equity. Circular economy is a **component** of green economy.
- Both link strongly to **SDG 12** (Responsible Consumption and Production).

Feature	Green economy	Circular economy
Focus	Overall low-carbon inclusive growth	Material/waste loop design
Scope	Broader	Narrower — subset of green
Example	Solar parks, green hydrogen	EPR, plastic recycling
SDG link	7, 8, 13 and more	Mainly SDG 12

15.5 Sustainable Agriculture & NMSA

Identity: Land quality intact | Mixed cropping strategy | NMSA under NAPCC

- **Sustainable agriculture** means producing food, fibre, and related products with farming techniques that protect the environment, public health, communities, and animal welfare.
- In short form, it means utilising land so that its **quality remains intact**.
- Heavy chemical fertilisers and pesticides have damaged soil health, which is why sustainable pathways emphasise nutrient management and ecological farming.
- An environment-friendly strategy set for Indian agriculture is **mixed cropping**, **organic manures**, **nitrogen-fixing plants**, and **pest-resistant crop varieties**.
- **Sikkim** became India's (and often cited as the world's) first **100% organic state** (process from **2003**; declared around **2016**) — NPOP is under Commerce/APEDA, not Rural Development.
- Expansion of cultivable land with heavy urea/superphosphate use, monoculture, and intensive aerial pesticide spray is **not** the sustainable strategy option.
- The **National Mission for Sustainable Agriculture (NMSA)** is a mission under the **National Action Plan on Climate Change (जलवायु परिवर्तन) (NAPCC)**.
- NMSA focuses on agricultural productivity, **water-use efficiency**, nutrient management, and **livelihood diversification** through sustainable pathways.
- **Conservation Agriculture** (FAO framing) rests on **minimum tillage**, **permanent soil cover with crop residues**, and **crop rotation / sequencing**. Avoiding plantation crops is **not** a Conservation Agriculture principle.

Idea	Correct fact
Sustainable agriculture	Land quality remains intact
Best strategy set	Mixed cropping + organic + N-fixing + pest-resistant varieties
NMSA	NAPCC mission
Conservation Agriculture	Min tillage + residue cover + rotation